04

How the Estimation Works, and Its Limitations

From a photo to rupees: how parts become square feet, square feet become material quantities with wastage, and quantities become material and labour costs at local rates — with every formula, a worked example checked against the test suite, and an honest account of where the numbers can be wrong.

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CODE
backend/core/estimation.py · managers/rates.py

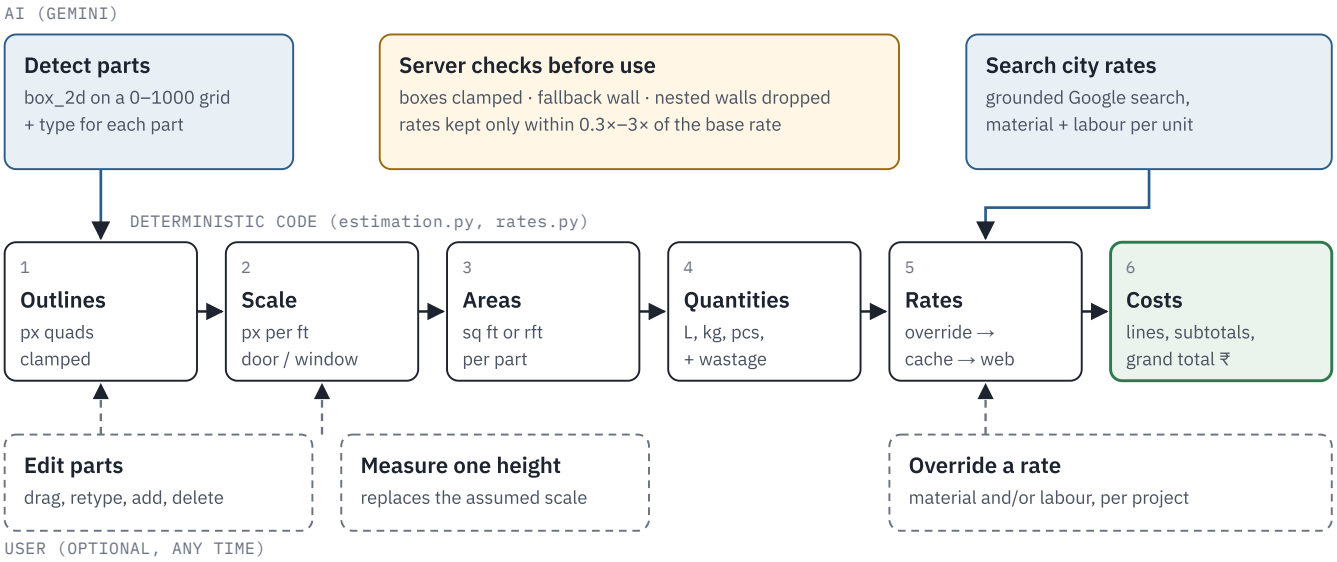
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STATUS
Advisory estimate, not a quotation

1 · PIPELINE

The estimation pipeline

The single most important design decision: **the AI never calculates a number that is priced.** Gemini locates parts and looks up published rates; every area, quantity and rupee is computed by deterministic Python that is covered by unit tests, so the same inputs always give the same estimate and every line can be checked by hand.



Material choices (from the design) select which catalog item — coverage, coats, wastage, unit — applies to each part.

Fig 1.1 Who computes what. Blue = AI suggestions; black = deterministic code; dashed = optional user corrections that feed straight into the maths.

2 · OUTLINES

From detection boxes to outlines

Gemini returns each part as `box_2d = [ymin, xmin, ymax, xmax]` on a normalised 0–1000 grid, independent of photo size. Each value is clamped to 0–1000 and scaled to the stored photo's pixels, giving a four-corner outline (a "quad"). Parts the user drags or adds are stored the same way, clamped to the image.

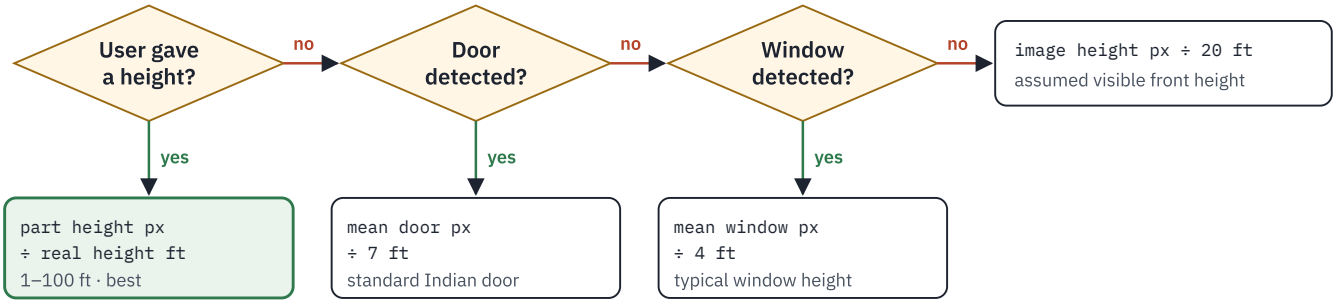
```
x = xmin / 1000 × photo_width        y = ymin / 1000 × photo_height
quad = [[x0, y0], [x1, y0], [x1, y1], [x0, y1]]
```

Cleanup before measuring: if no wall was detected, a fallback wall covering all detected parts is added (so a facade is always priceable); a wall lying more than 80 % inside a larger wall is dropped (so the same surface is not priced twice); any remaining wall overlap above 20 % is flagged on the estimate.

3 · SCALE

Scale: pixels per foot

A photo has no units. One scale converts pixels to feet for the whole photo, chosen in this order:



The scale is recomputed whenever parts change, until the user enters a real measurement; after that it is kept. The estimate always states which source was used, and the PDF repeats it as an assumption.

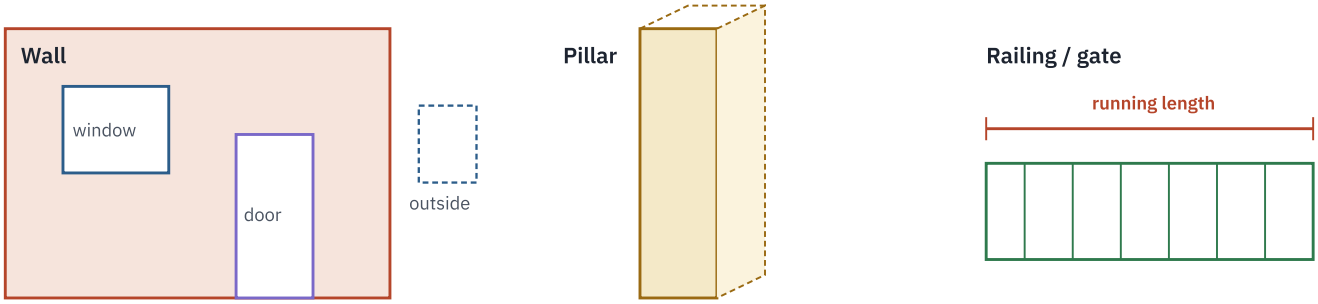
Fig 3.1 Scale selection (`default_scale`, `scale_from_component`).

4 · AREAS

Areas and lengths

The pixel area of each outline is computed with the shoelace formula, then converted with the scale. Outlines are stored as four points, so a user can skew a box to follow a sloped edge.

```
area_px   = ½ · | Σ (xᵢ · yᵢ₊₁ − xᵢ₊₁ · yᵢ) |        (shoelace, over the 4 corners)
area_sqft = area_px ÷ px_per_ft²
length_rft = outline width px ÷ px_per_ft          (for materials priced per running foot)
```



net = wall − openings whose centre is inside it (never below 0)

visible face × 4: treated as a square column, only one face shows

rft materials: length = box width; height does not change the price

Fig 4.1 Part-specific rules (`measure()`). Balconies, parapets, gates (for sq ft materials) and roof edges use their area as drawn; windows and doors are never priced.

All parts that share a material are summed (e.g. 4 walls → one paint line covering WALL×4), and the total is rounded to 0.1 before quantities and costs are worked out. Parts the user marked “leave as it is” are skipped.

5 · QUANTITIES

Quantities with wastage

Each catalog material carries its own pricing unit, coverage and wastage. `factor = 1 + wastage % / 100`.

Unit bought in	Formula	Materials
Litres (L) or kg	`measure × coats ÷ coverage_per_unit × factor`, 1 decimal	Exterior emulsion (2 coats, 110 sq ft/L), elastomeric (2 coats, 90 sq ft/L), texture (1 coat, 10 sq ft/kg)
Pieces (pcs)	`ceil(measure ÷ piece_sqft × factor)` — whole tiles	Vitrified tiles (2 sq ft each), brick cladding tiles (0.1875 sq ft each)
sq ft or rft	`measure × factor`, 1 decimal	Stone cladding, ACP and WPC panels (sq ft); glass, SS and MS railing (rft)

5 · CATALOG

Catalog parameters used

Material	Category	Priced per	Bought in	Coats / coverage	Wastage	Base ₹ mat + lab	Lasts
Exterior emulsion paint	Paint	sq ft	L	2 · 110 sq ft/L	5 %	16 + 10	5–7 yrs
Elastomeric waterproof paint	Paint	sq ft	L	2 · 90 sq ft/L	5 %	28 + 12	8–10 yrs
Texture finish	Texture	sq ft	kg	1 · 10 sq ft/kg	8 %	35 + 22	8–10 yrs
Natural stone cladding	Stone	sq ft	sq ft	—	10 %	120 + 60	25+ yrs
Slate stone cladding	Stone	sq ft	sq ft	—	10 %	90 + 55	25+ yrs
Vitrified elevation tiles	Tiles	sq ft	pcs (2 sq ft)	—	10 %	70 + 40	15–20 yrs
Brick cladding tiles	Tiles	sq ft	pcs (0.19 sq ft)	—	12 %	85 + 45	20+ yrs
ACP panels (4 mm)	Panels	sq ft	sq ft	—	8 %	180 + 70	15–20 yrs
WPC wood-look louvers	Panels	sq ft	sq ft	—	8 %	260 + 80	15–20 yrs
Toughened glass railing	Railing	rft	rft	—	5 %	1,800 + 400	20+ yrs
Stainless steel railing	Railing	rft	rft	—	5 %	1,100 + 300	20+ yrs
Mild steel railing (painted)	Railing	rft	rft	—	5 %	550 + 200	8–12 yrs

Base rates are the national reference (tier-2 city) used for sanity bounds and as the last fallback; real estimates use city rates (§7).

6 · COSTS

Costs and totals

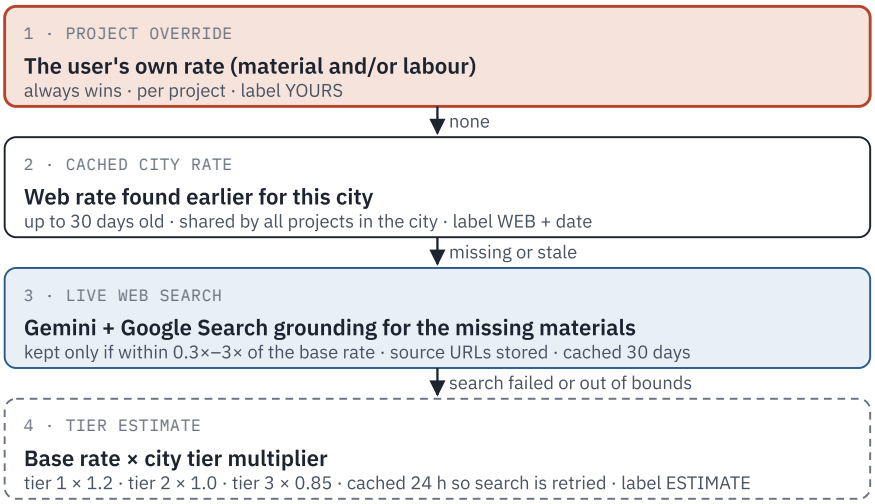
```
material_cost = measure × (1 + wastage) × material_rate      → rounded to ₹1
labour_cost   = measure × labour_rate                        → rounded to ₹1
line_total    = material_cost + labour_cost
category subtotal = Σ line totals in the category (Paint, Panels, Railing, …)
grand total       = Σ all line totals
```

Wastage applies to the material bought, not to labour: a painter is paid for the area painted, not for the paint spilled. Each line also shows its rate source (WEB with the date it was fetched, ESTIMATE, or YOURS) and the parts it covers.

7 · RATES

Where the rates come from

Rates are resolved per material for the project's state and city, highest priority first:



Why the sanity bounds

A search can return a price per box, per kg or per square metre, or an outlier. Rejecting anything below 0.3× or above 3× of a known reference stops a single bad result from inflating an estimate tenfold.

Example (Ludhiana)

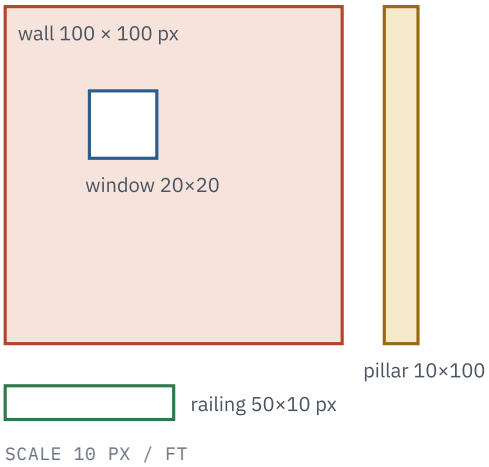
WPC: web ₹210 vs base ₹260 → kept
MS railing: web ₹300 vs ₹550 → kept (both within 0.3×–3×)

Fig 7.1 Rate resolution (`managers/rates.py`). The 91 cities across 24 states and union territories are assigned tiers 1–3.

8 · WORKED EXAMPLE

Worked example (from the tests)

This exact case is asserted in `backend/core/tests/test_estimate.py`, so the numbers below are what the code produces.



Step	Working	Result
Wall	10,000 px² ÷ 10² = 100; window 400 ÷ 100 = 4	96 sq ft
Pillar	1,000 ÷ 100 = 10 × 4 faces	40 sq ft
Paint measure	96 + 40	136 sq ft
Paint quantity	136 × 2 ÷ 110 × 1.05	2.6 L
Paint material	136 × 1.05 × ₹15	₹2,142
Paint labour	136 × ₹10	₹1,360
Paint line	2,142 + 1,360	₹3,502
Railing	50 ÷ 10 = 5 rft; 5 × 1.05 × ₹1,000 + 5 × ₹200	₹6,250
Grand total	**3,502 + 6,250**	**₹9,752**

9 · REAL EXAMPLE

Real example: Ludhiana house

The live demonstration house (Document 03): 11 detected parts, no user measurement, so the scale came from the detected door: **door height in px ÷ 7 ft = 25.19 px/ft**. Ludhiana web rates found on 31 Oct 2026.

Line	Parts	Measure	Quantity working	Material working	Labour working	Total
Exterior emulsion	WALL×4	700.9 sqft	700.9×2÷110×1.05 = 13.4 L	700.9×1.05×15 = ₹11,039	700.9×12 = ₹8,411	₹19,450
WPC louvers	BALCONY×2	459.6 sqft	459.6×1.08 = 496.4 sqft	459.6×1.08×210 = ₹1,04,237	459.6×65 = ₹29,874	₹1,34,111
ACP panels	ROOF_EDGE×1	138.7 sqft	138.7×1.08 = 149.8 sqft	138.7×1.08×150 = ₹22,469	138.7×120 = ₹16,644	₹39,113
MS railing	GATE×1	15.6 rft	15.6×1.05 = 16.4 rft	15.6×1.05×300 = ₹4,914	15.6×125 = ₹1,950	₹6,864
Grand total (Panels ₹1,73,224 · Paint ₹19,450 · Railing ₹6,864)				**₹1,42,659**	**₹56,879**	**₹1,99,538**

10 · ACCURACY

Accuracy and sensitivity

The largest source of error is the **scale**, because area depends on its square:

```
area ∝ 1 / px_per_ft²
scale off by +10 % → areas −17 %
scale off by −10 % → areas +23 %
```

A door that is really 7.5 ft instead of the assumed 7 ft makes every area about 13 % too small. Entering one measured height removes this error entirely, which is why the app offers it on the Parts step and the estimate says which scale it used.

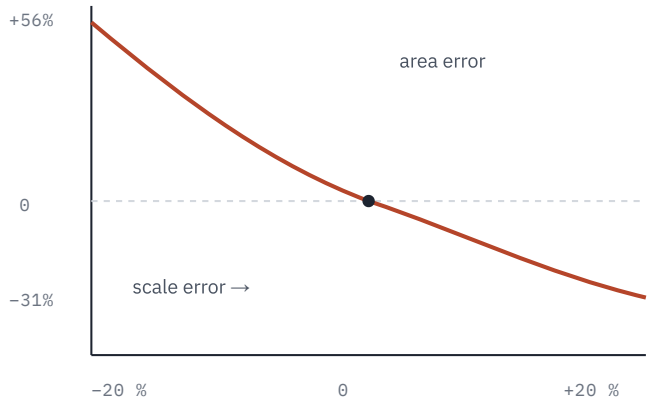


Fig 10.1 Area error versus scale error (area = 1/(1+e)² − 1).

Source of error	Typical effect	How Miesign limits it
Assumed scale	±10–25 % on every area	One measured height replaces it; the scale source is shown on the estimate and in the PDF
Perspective (photo at an angle)	Near parts look larger than far ones	Upload guidance asks for a straight-on photo from across the road
Box outlines	Over-covers sloped or irregular surfaces	Corners can be dragged into any four-sided shape
Missed or duplicate parts	Area missing or counted twice	Fallback wall, nested-wall removal, overlap warning, full manual editing
Rates	Market spread between suppliers	City rates with sources, sanity bounds, and per-project overrides

11 · LIMITATIONS

Limitations of the system

Miesign gives an **advisory** estimate from one photo, as the brief allows (§7: estimates are advisory, not legally binding). These are the known limits, how much they matter, and what the system already does about them.

Limitation	Impact	Detail	Current mitigation
Measurement			
One visible facade only	High	Sides and back of the house are not measured, so a full repaint is under-estimated.	Stated in every estimate and PDF; more photos can be uploaded and one chosen.
No depth information	Medium	One scale for the whole photo; an angled photo stretches near parts.	Upload tips ask for a straight-on view.
Assumed scale	Medium	7 ft door / 4 ft window / 20 ft front when nothing is measured.	Optional measured height; scale source shown.
Rectangular detection	Medium	Boxes over-cover angled or irregular parts; Gemini can miss or duplicate parts.	All parts editable; fallback wall; nested walls dropped; overlap warning.
Preview			
Generative render	Medium	The image model can alter small details (seen once: an extra railing) or, before the fix in Doc 03 §5, skip a surface.	Hard rules in the prompt, part locations given, swatch card; “treat it as a guide” shown on screen.
Colour under lighting	Low	Hue within 2–4° of the request on real photos, but about 70 % of saturation kept; renders look slightly duller than the swatch.	Exact hex in the prompt plus a colour card; named shades shown with hex.
Render cap	Low	8 previews per project to control AI cost.	Unchanged designs reuse renders; failed renders are refunded.
Rates and costs			
Rates are averages, not quotes	Medium	Web search returns typical published prices; local suppliers differ.	Sanity bounds, source links, labelled source, user overrides.
Scope of costs	Medium	No scaffolding, surface repair, primer/putty as separate items, transport, taxes or contractor margin.	Labelled advisory; overrides let a contractor fold these into rates.
Product and infrastructure			
No accounts	Medium	Anyone with a project link can view and edit it.	Unguessable UUID links; documented.
Single small server	Low	SQLite and local files on one e2-micro; fine for a prototype, not for heavy traffic.	8 concurrent requests; SQLite lock wait; budget alert.
Language and currency	Low	English and INR only; 91 Indian cities.	—
Building type	Low	Designed for low-rise residential fronts (brief §7); high-rise or commercial photos are rejected or measured poorly.	Photo check rejects non-house photos.

12 · NEXT

What would improve it

- **Depth and perspective:** a monocular depth model or two photos to correct perspective and measure side walls.
- **Segmentation masks** instead of boxes, for exact areas of irregular surfaces.
- **Automatic scale** from several references (doors, windows, floor heights) cross-checked against each other.
- **Supplier rate feeds** and contractor-submitted quotes per city instead of web averages.
- **Full cost scope:** surface preparation, scaffolding, GST and margin as optional lines.
- **Accounts and sharing** with view-only links; Postgres and object storage for scale.